Artificial Intelligence

Local search algorithms (AIMA sections 4.1)

Summary

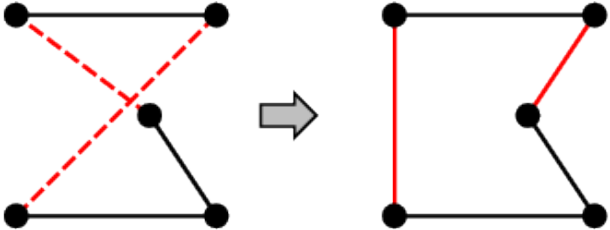
- Hill-climbing
- Simulated annealing
- Genetic algorithms (briefly)

Iterative improvement algorithms

- In many optimization problems, **path** is irrelevant: the goal state is the solution
- Then state space = set of "complete" configurations;
  - find **optimal** configuration, e.g., TSP, etc.
  - find configuration satisfying constraints, e.g., n-Queens
- Can use iterative improvement algorithms
  - keep a single "current" state
  - try to improve it
- Constant space, suitable for online as well as offline search

Example: Travelling Salesperson Problem

Start with any complete tour, perform pairwise exchanges



Variants of this approach get within 1% of optimal very quickly with thousands of cities

Example: n-queens

◇ Put n queens on an $n \times n$ board with no two queens on the same, row, column, or diagonal

◇ Move a queen to reduce number of conflicts

h = 5 h = 2 h = 0

Almost always solves n-queens problems almost instantaneously for very large n, e.g., $n = 1 million$

Hill-climbing (or gradient ascent/descent)

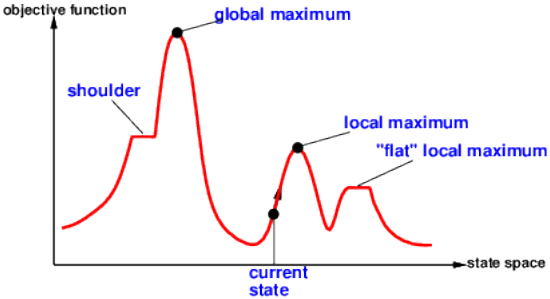
"Like climbing Everest in thick fog with amnesia"

```
function Hill-Climbing( problem) returns a state that is a local maximum
    inputs: problem, a problem
    local variables: current, a node
                     neighbor, a node

    current ← Make-Node(problem.Initial-State)
    loop do
        neighbor ← a highest-valued successor of current
        if neighbour.Value ≤ current.Value then return
            current.State
        end if
        current ← neighbor
    end
```

Hill-climbing contd.

Useful to consider state space landscape



Random-restart hill climbing overcomes local maxima—trivially complete
Random sideways moves escape from shoulders loop on flat maxima

Simulated Annealing

- Inspired by statistical mechanics
- Idea: escape local maxima by allowing some "bad" moves, **but gradually decrease their frequency**
- Allow more random moves at the beginning
  - we can reach zones with better solutions
- Diminish probability of having a random move towards the end
  - refine search around a good solution

Simulated annealing (pseudo-code)

```
function Simulated-Annealing( problem, schedule) returns a solution state
   inputs: problem, a problem
           schedule, a mapping from time to "temperature"
   local variables: current, a node
                    next, a node
                    T, a "temperature" controlling prob. of downward steps

   current ← Make-Node(problem.Initial-State)
   for t ← 1 to ∞ do
       T ← schedule(t)
       if T = 0 then return current
       next ← a randomly selected successor of current
       ΔE ← next.Value − current.Value
       if ΔE > 0 then current ← next
       else current ← next only with probability e^(Δ E/T)
```

Properties of simulated annealing

At fixed "temperature" T, state occupation probability reaches Boltzman distribution

$$p(x) = \alpha e^{\frac{E(x)}{kT}}$$

T decreased slowly enough $\Longrightarrow$ always reach best state x^*
because $e^{\frac{E(x^*)}{kT}} / e^{\frac{E(x)}{kT}} = e^{\frac{E(x^*)-E(x)}{kT}} \gg 1$ for small T
Is this necessarily an interesting guarantee??

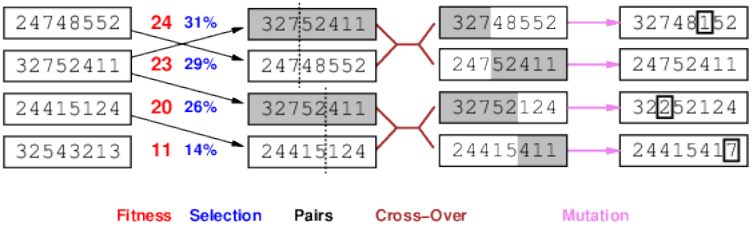
- ◇ Devised by Metropolis et al., 1953, for physical process modelling
- ◇ Widely used in VLSI layout, airline scheduling, etc.

Local beam search

Idea: keep k states instead of 1; choose top k of all their successors
Not the same as k searches run in parallel!
Searches that find good states recruit other searches to join them
Problem: quite often, all k states end up on same local hill
Idea: choose k successors randomly, biased towards good ones
Observe the close analogy to natural selection!

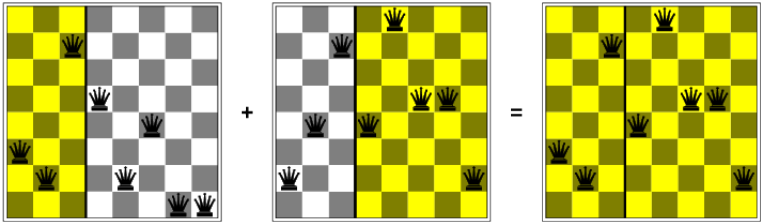
Genetic algorithms

= stochastic local beam search + generate successors from **pairs** of states



Genetic algorithms contd.

GAs require states encoded as strings (GPs use programs)
Crossover helps **iff substrings are meaningful components**



GAs $\neq$ evolution: e.g., real genes encode replication machinery!

Exercise: Local Search for the 4-Queens problem

Consider the 4-Queens problem. Assume the evaluation function is the number of pairs of queens that attack each other. Assume initial state is (1234)

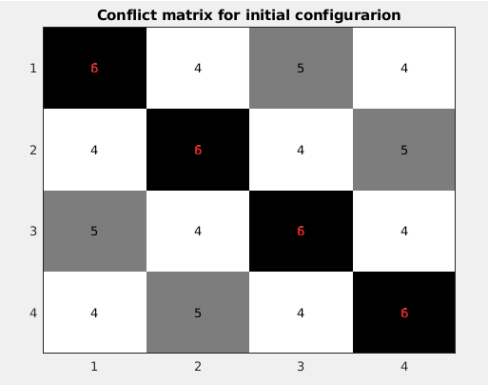
- What is the current score for the initial state
- Write down the values of all successor states for this initial state
- Implement a simple program that computes the next best state(s) for your hill-climbing approach
- Trace a possible execution of a (deterministic) hill-climbing approach
- Comment on optimality of final state
- **sol**: eserc1LocalSearch.m

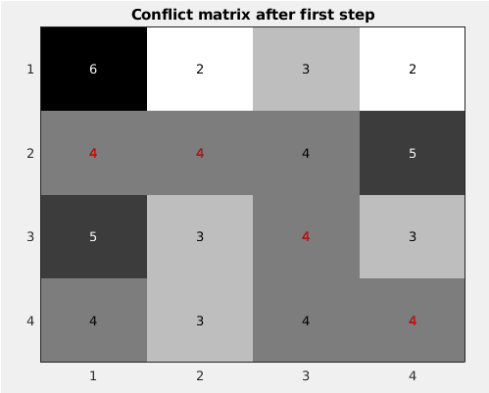
Exercise: Local Beam Search for the 4-Queens problem

Consider the 4-Queens problem and the deterministic hill climbing approach described above. Assume $k = 3$ and initial states are: $(1234), (2222), (3333)$.

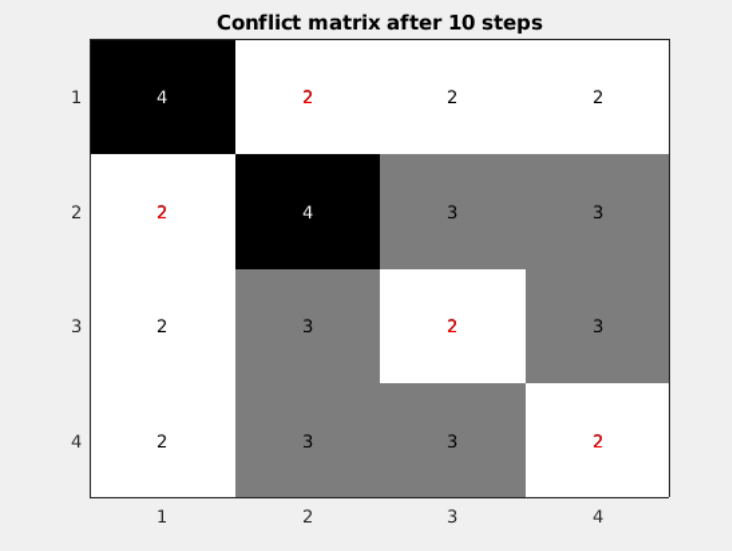
- Trace execution of a **parallel** search
  sol: eserc2Parallel.m
- Trace the execution of a **beam** search
  sol: eserc2BeamSearch.m

Local Search for the 4-Queens problem – solution 1/2

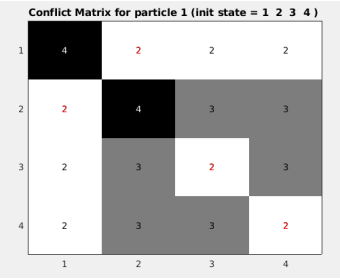


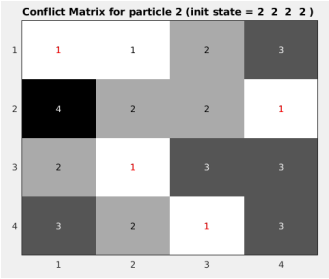


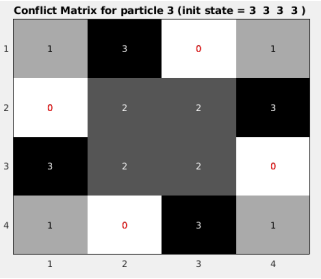
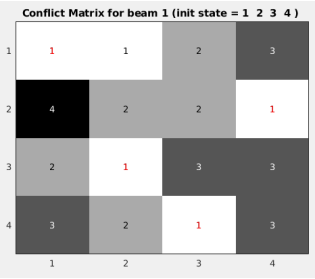
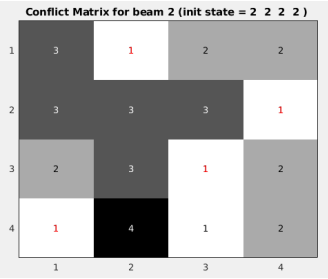
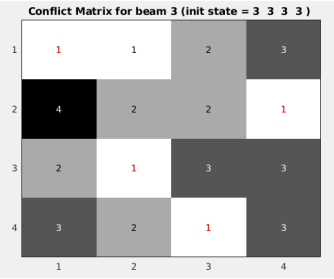
Local Search for the 4-Queens problem – solution 2/2



Parallel Search for the 4-Queens problem – solution







Beam Search for the 4-Queens problem – solution